CORDEX-ML-Bench — Overall Rank Summary

GENERATIVE

	Precipitation (pr)		Max. Temp. (tasmax)	
	pr ESD	**pr** Emulator	**tasmax** ESD	**tasmax** Emulator
	▲ CV ▽ Extrap.	▲ CV ▽ Interp.	▲ CV ▽ Extrap.	▲ CV ▽ Interp.
Generative Models				
RCMGEM-mv-orog	1 / 1	1 / 1	1 / 1	2 / 2
RCMFlow-orog	2 / 3	2 / 2	2 / 15	25 / 25
ViT-IFCAv1	4 / 2	5 / 6	3 / 29	6 / 6
ViT-IFCAv1-orog	5 / 6	4 / 5	6 / 25	8 / 8
FlowMatching-v1	10 / 7	10 / 10	7 / 4	5 / 7
FlowMatching-v1-orog	6 / 5	9 / 11	18 / 5	7 / 10
ResGAN-v2	8 / 9	12 / 9	11 / 17	22 / 19
ResGAN-v2-orog	3 / 4	8 / 8	8 / 13	23 / 23
ResGAN	9 / 11	7 / 7	9 / 27	26 / 21
ResGAN-orog	7 / 8	6 / 4	13 / 24	27 / 18
EnScale	11 / 12	14 / 13	10 / 20	16 / 22
EnScale-orog	14 / 16	13 / 14	15 / 22	20 / 27
EnScale-linex	12 / 15	16 / 16	5 / 3	10 / 13
ParamDiffusion-orog	16 / 14	3 / 3	14 / 14	1 / 1
UiBCorrDiff-orog	21 / 29	36 / 35	12 / 11	12 / 16
CorrDiff-TW1_NTNU-CWA	28 / 24	21 / 20	33 / 19	4 / 3
CorrDiff-TW1_NTNU-CWA-orog	29 / 25	19 / 21	32 / 21	3 / 4
spaGAN	38 / 35	33 / 29	35 / 37	35 / 34
spaGAN-orog	34 / 31	29 / 26	34 / 35	34 / 30

DETERMINISTIC

	Precipitation (pr)		Max. Temp. (tasmax)	
	pr ESD	**pr** Emulator	**tasmax** ESD	**tasmax** Emulator
	▲ CV ▽ Extrap.	▲ CV ▽ Interp.	▲ CV ▽ Extrap.	▲ CV ▽ Interp.
Deterministic Models				
DeepESDcrps_IFCAv2	13 / 13	11 / 12	36 / 38	36 / 35
DeepESDcrps_IFCAv2-orog	15 / 10	22 / 23	29 / 10	31 / 31
DeepESD_IDL	19 / 22	25 / 24	40 / 40	40 / 40
DeepESD-IFCAv1	23 / 21	17 / 17	26 / 34	17 / 26
DeepESD-IFCAv1-orog	24 / 19	18 / 19	17 / 2	11 / 14
CNRM-UNeT	25 / 23	28 / 30	22 / 23	13 / 12
DetUNet	22 / 27	20 / 18	16 / 28	29 / 24
DetUNet-orog	18 / 20	15 / 15	19 / 26	28 / 15
ParamUNET	27 / 26	31 / 33	24 / 16	18 / 28
DetUNet-v2	26 / 28	26 / 27	21 / 18	21 / 17
DetUNet-v2-orog	20 / 18	23 / 22	4 / 9	15 / 11
GNN4CD-orog	31 / 30	34 / 31	28 / 8	24 / 29
Prithvi-UNet	33 / 33	38 / 38	39 / 32	37 / 36
Prithvi-UNet-orog	17 / 17	30 / 32	25 / 7	30 / 32
Rossby-UNet	32 / 36	27 / 28	27 / 36	14 / 9
Rossby-UNet-orog	30 / 37	24 / 25	20 / 33	9 / 5
ANN	36 / 32	35 / 36	37 / 39	38 / 37
ANN-orog	37 / 34	37 / 37	30 / 6	32 / 33
XGBoost_IDL	35 / 39	32 / 34	23 / 30	19 / 20
DeepSensor	40 / 40	40 / 40	38 / 31	39 / 38
DeepSensor-orog	39 / 38	39 / 39	31 / 12	33 / 39

Overall rank

1 (best) — 40 (worst)

▲ upper-left = Cross-validation rank

▽ lower-right = Extrapolation / Interpolation rank

Rank across 40 models with data · method='first': ties broken by list order · RMSE = sqrt(mean(x^2))